MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) FIFTH SEMESTER EXAMINATION

MG5DSECSC300- COMPUTER SECURITY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Max. Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer **all** questions

Each question carries **2** marks

1) Describe the importance of security in a computer network. [U] [CO1]
2) State the purpose of password policies. [U] [CO3]
3) Identify the three basic factors used for authentication. [U] [CO3]
4) Define biometric authentication. [U] [CO4]
5) Identify which VPN architecture is suitable for securely connecting two branch networks. [U] [CO4]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6) Explain the major approaches used for computer security. [K] [CO1]
7) Differentiate between specific security mechanisms and pervasive security mechanisms. [U] [CO1]
8) Compare active attacks and passive attacks with suitable examples. [U] [CO1]

9) Discuss the importance of security awareness training in reducing the success rate of phishing attacks. [U] [CO1]

10) Encrypt the message "HELLO" using a Caesar cipher with a shift of 3 and show the steps. [A] [CO2]

11) Discuss the advantages and limitations of the Vigenère cipher. [U] [CO2]

12) Explain Network Behavior Analysis (NBA) and its significance in intrusion detection. [U] [CO3]

13) Explain packet-filtering firewalls and how they use packet information to make filtering decisions. [U] [CO4]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14) Explain monoalphabetic cipher in detail? Describe its encryption mechanism, illustrate it with a suitable example, and discuss its limitations and vulnerability to frequency analysis. [A] [CO2]

15) Explain various techniques used by malware to infect and compromise computer systems. [U] [CO3]

16) Define biometric authentication. Examine the various biometric techniques employed in security systems, highlighting their advantages and limitations. [U] [CO4]